

Markout Analysis — Hyperliquid Wallets (BTC / ETH / SOL / HYPE)

Post-fill markout study of Hyperliquid taker wallets. We measure per-entry markout across five horizons and four coins, rank wallets by markout, test whether top-ranked wallets persist out-of-sample, and characterize the subset that does. Every out-of-sample number below is chronologically split (train → embargo → test), and figures are entry-weighted unless labeled otherwise.

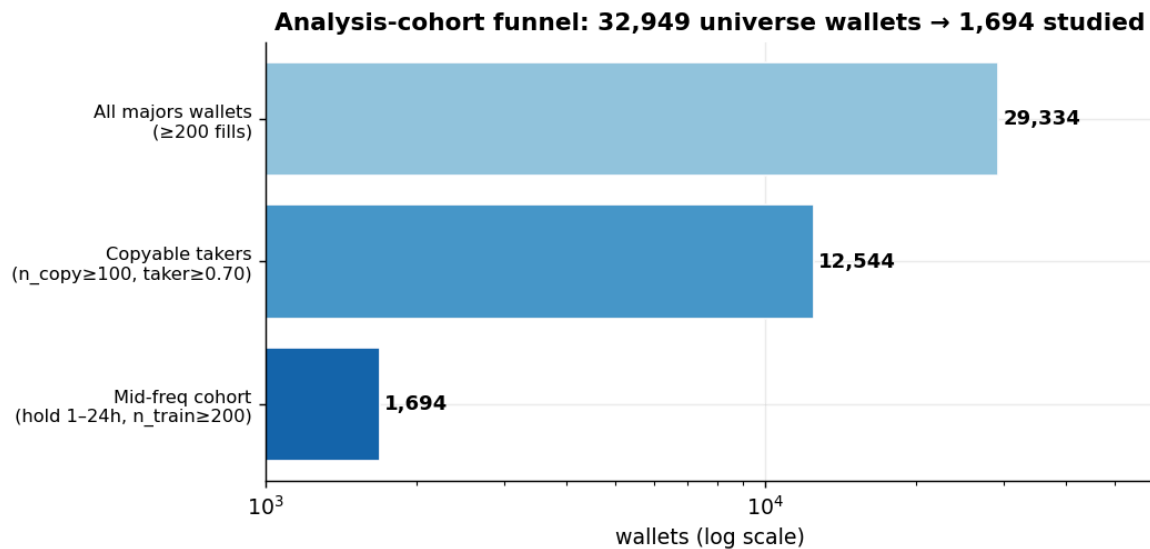
Summary of findings

1. **Markout term structure is small and coin-shaped.** Field markout is within ± 1 bp at 1–4h for BTC/ETH/SOL and bleeds to -6 to -8 bp by 24h. HYPE is the exception — a positive hump peaking $\sim +3$ bp at 8h — but that hump is almost entirely coin drift (neutralized HYPE 24h is -18.6 bp vs raw $+2.8$ bp).
 2. **Top in-sample markout wallets look excellent** ($+40$ to $+130$ bp gross at 8h, 60–76% hit rates) **and do not persist out-of-sample.** Train→test rank-IC is $+0.017$ ($p=0.59$); selecting on best in-sample markout returns **-28.7 bp net OOS** (winner's curse). This holds across every ranking metric we tried (Sharpe, Sortino, t-stat, shrinkage, PnL, hit-rate, ...).
 3. **A weak-but-real ranking signal survives** at the population/decile level (monthly rank-IC $+0.093$ at 8h, 10/10 folds; top-decile forward $+4.5$ bp over field) — but it is **generic short-horizon reversal**, not wallet skill: the top decile's entire edge is captured by a costless mechanical fade (alpha over fade = $+1.0$ bp, $p=0.44$).
 4. **The wallets that rank consistently are contrarian faders.** We isolate a frozen 121-wallet recurring cohort; 83% are net faders and they enter hard *against* large, decelerating moves at deep pullbacks. Their edge is the fade, done harder — not private information. Adding wallet identity/direction to a market-state model improves 8h-return prediction by **-0.0003 IC** (i.e., nothing).
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1. Data & universe

- **Raw universe:** 7,332,013 taker entries from **32,949 wallets** across 273 coins, 2025-08-01 → 2026-06-30 (UTC). Per coin: BTC 2.13M entries / 27.1k wallets, ETH 1.04M / 24.1k, SOL 790k / 22.0k, HYPE 771k / 19.9k.
- **Analysis cohort:** to study *copyable* behavior we freeze a mid-frequency taker cohort — wallets with ≥ 200 majors fills, taker-share ≥ 0.70 , median hold 1–24h, and ≥ 200 training decisions. This yields **1,694 wallets / 812,616 priced entries** on the four majors.
- **Splits (chronological):** TRAIN $< 2026-02-01$ (583,032 entries) · EMBARGO Feb-2026 (61,116, discarded) · TEST 2026-03-01 → 06-30 (168,468). The test window is a -10.85% BTC down-market.
- **Markout definition:** $\text{raw_Xh} = \text{dir} \cdot (\text{px}(\text{entry} + \text{X}) / \text{px_entry} - 1) \cdot 1\text{e4}$ (bp). Positive = price moved in the trade's direction after the fill. Horizons: 1h / 2h / 4h / 8h / 24h. Every trade is

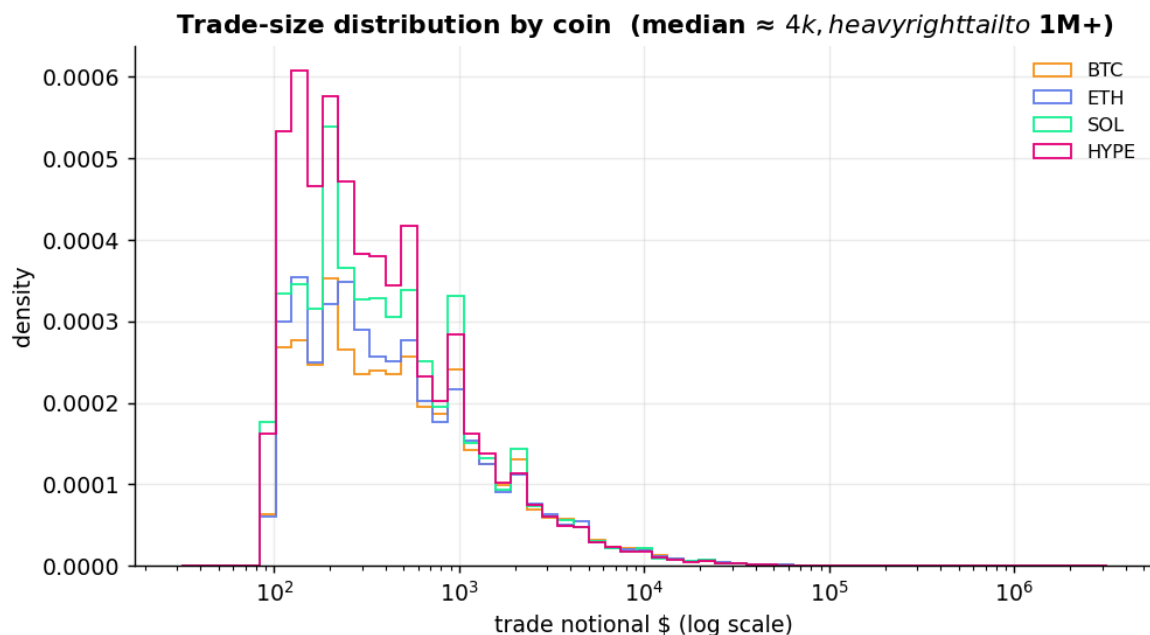
evaluated as if exited at the horizon, so the horizon markout *is* the trade's PnL.



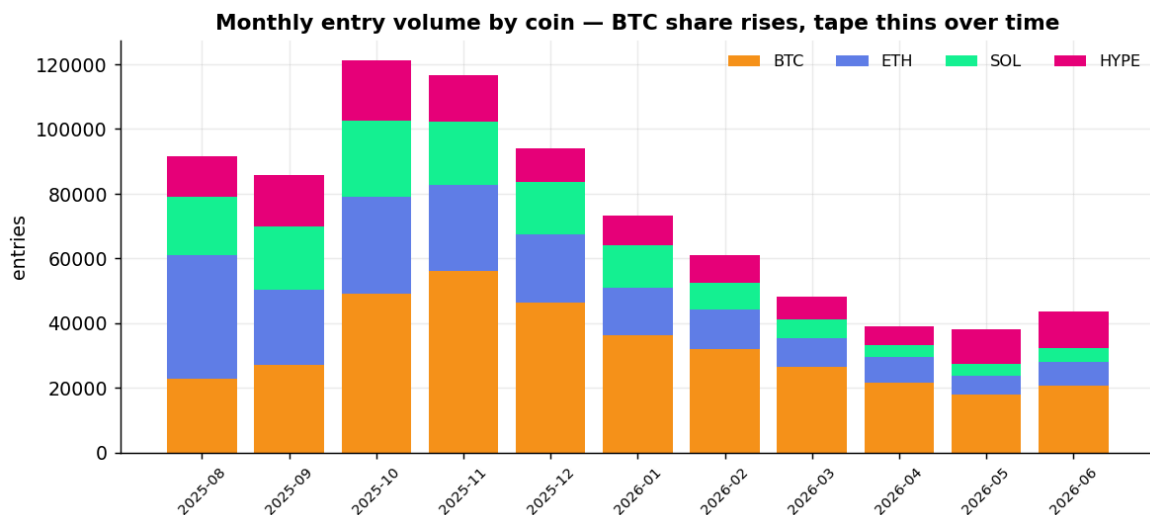
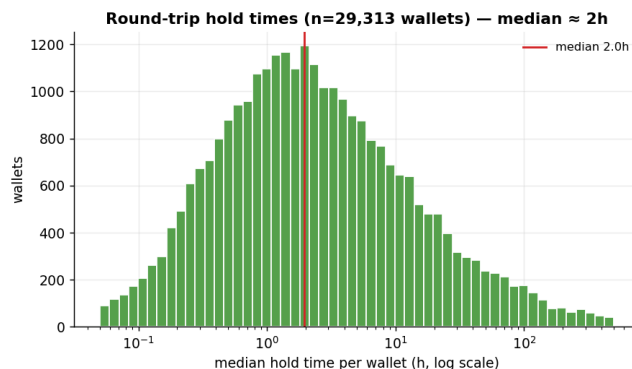
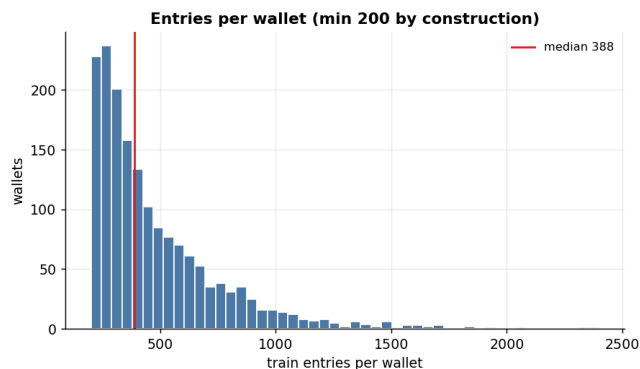
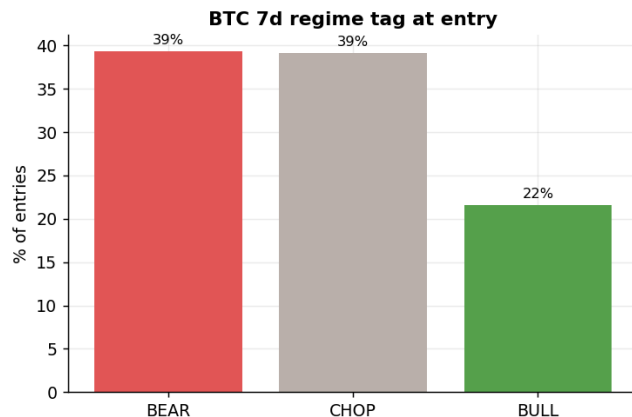
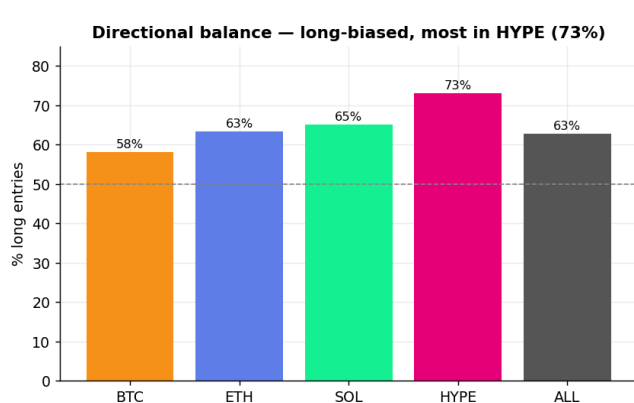
Cohort funnel

2. Wallet EDA

The cohort is long-biased (63% of entries long, rising to 73% in HYPE), trades a median ~\$4k ticket with a heavy right tail past \$1M, holds a median ~2h round-trip, and is active a median 97 of ~330 days. Entry volume shifts toward BTC over the window (BTC's monthly entry share rises from 25% in Aug to a 55% peak in Apr, easing to 47% by Jun) while the overall tape thins — worth remembering when comparing early vs late months.



Trade size distribution

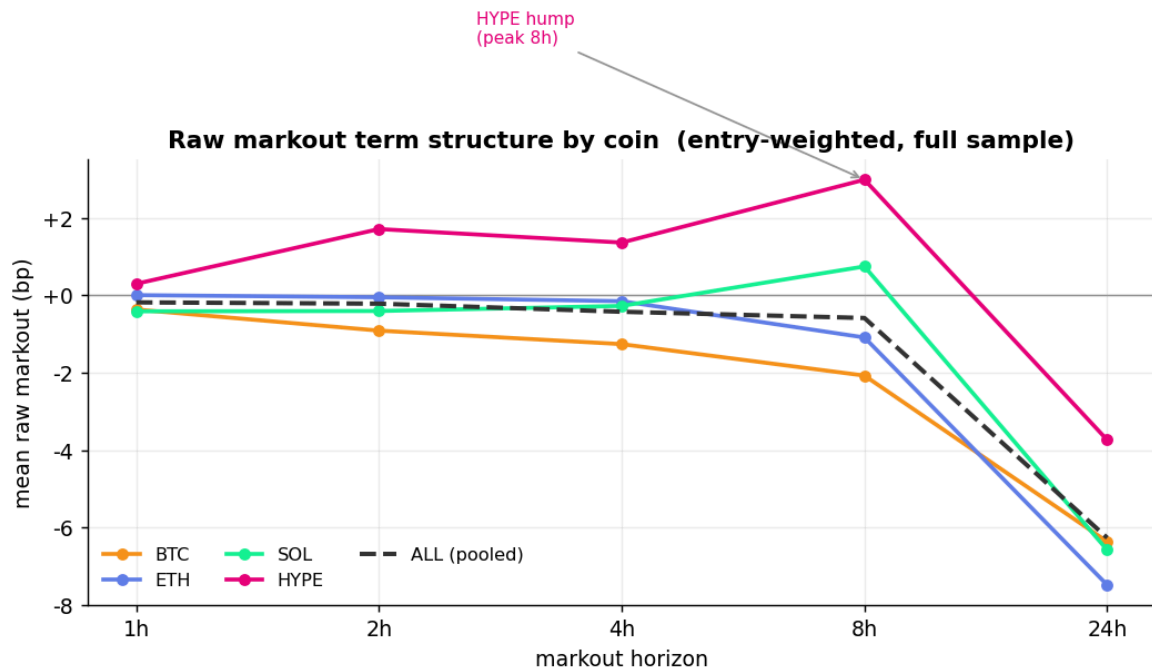


Temporal composition

Hold time is real round-trip data (29,334 wallets); trades are otherwise evaluated at the fixed markout horizon. Max-drawdown, where shown later, is computed on a non-overlapping ($\geq$ horizon-spaced) entry subset — a drawdown on the raw overlapping-horizon series is invalid (it scales with trade cadence, not risk).

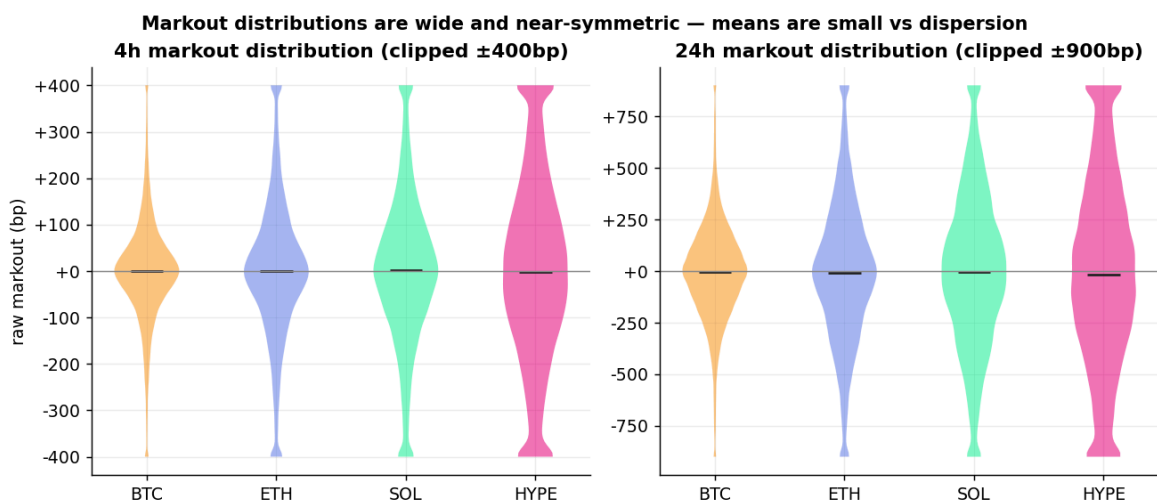
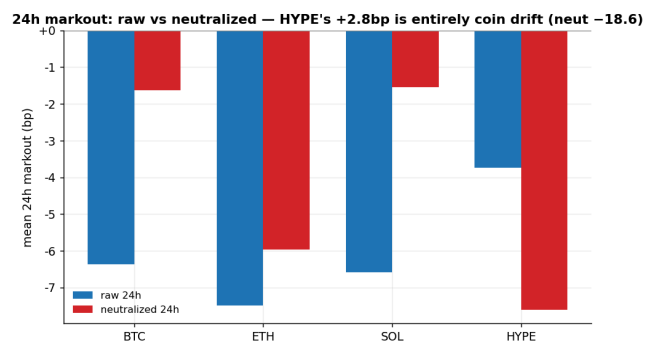
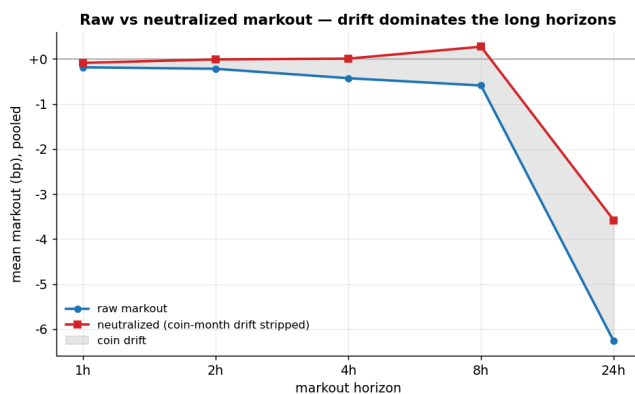
3. Markout term structure per coin per horizon

Entry-weighted field markout, full sample. BTC bleeds monotonically negative; ETH is flat through 4h then drops at 24h; SOL turns mildly positive at 4–8h before bleeding; **HYPE is the only clear positive hump, peaking ~+3 bp at 8h**. Dispersion grows with horizon (std 93 → 363 bp pooled; HYPE widest at every horizon) — the means are an order of magnitude smaller than the noise.



Raw term structure

The HYPE hump is **coin drift, not skill**. Stripping each coin's own monthly forward-return drift (neutralized markout) flattens the pooled curve and, for HYPE at 24h, flips +2.8 bp → -18.6 bp:



Distributions

Cross-checked against an independent whole-market taker field study ($N \approx 0.8\text{--}2.1\text{M}$ per coin): BTC bleed, ETH shape, and the HYPE 8h hump all reproduce.

4. Top markout wallets (in-sample)

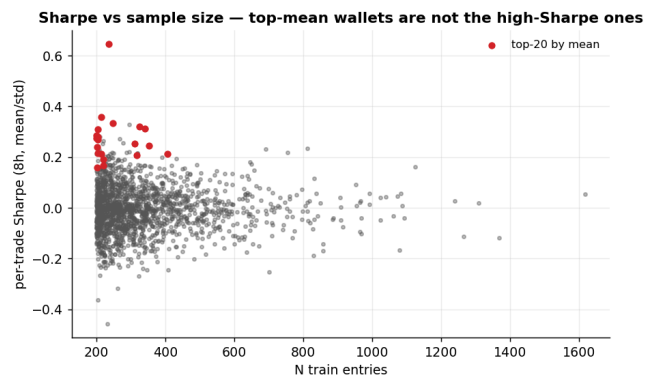
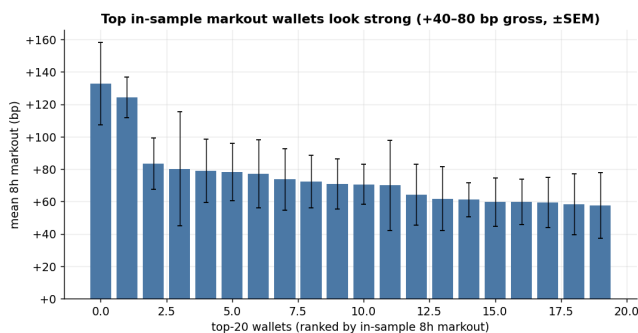
Ranked on TRAIN mean markout, min 100 entries (the cohort floor is already 200). Sharpe is per-trade (mean/std, un-annualized); MDD* is on the non-overlapping $\geq 8h$ -spaced subset; hit% is share of entries with positive markout. **These are in-sample snapshots — impressive, and (next section) not repeatable.**

Top 20 by 8h markout:

#	wallet	N	mean 8h (bp)	Sharpe	hit%	MDD* (bp)	med \$
1	0x5aadb434...	214	+132.8	0.36	69	1338	12,061
2	0xfd490cf8...	236	+124.2	0.65	76	1676	131,053
3	0xdfb34089...	248	+83.3	0.33	64	1400	5,535
4	0x172f5a11...	202	+80.2	0.16	51	2917	3,225
5	0x1ecc7c22...	200	+78.9	0.29	61	802	25,028
...							
20	0xcecc2959...	221	+57.6	0.19	60	926	493

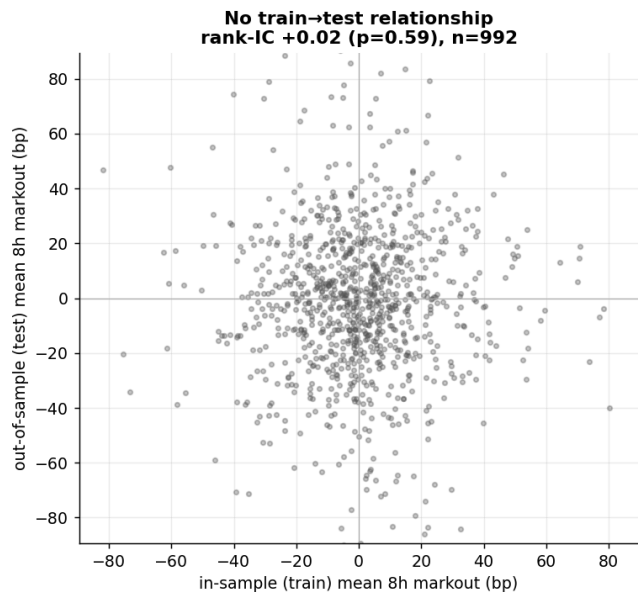
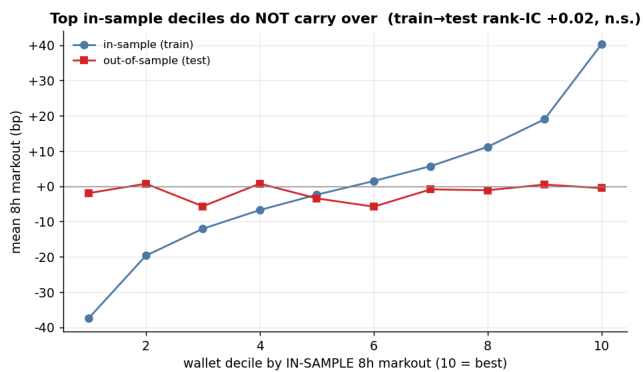
Top 5 by 1h markout (only 5 of 20 overlap with the 8h list — “top markout” is horizon-dependent):

0xfe104b72 +36.6 / 0x9899db38 +32.0 / 0xfffa6fc6 +29.8 / 0x413c2724 +26.0 / 0x032ea203 +23.2 bp.



5. Top wallets do not persist out-of-sample

This is the core negative result. Ranking wallets by in-sample 8h markout and measuring the same wallets in the test window: **the top deciles regress to the field.**



- **Train→test rank-IC = +0.017 (p=0.59)** — no relationship (992 wallets, ≥ 50 train / ≥ 15 test entries).
- **The exact top-20 basket** earns +3.75 bp gross [−7.6, +15.3] p=0.26 in test, collapsing to **+0.20 bp at realistic cost**; it depends on ≤ 2 wallets and flips sign every month.
- **Top-decile → top-decile transition = 1.09× base rate (p=0.22)** — barely above chance.
- **Selection cost:** picking wallets on best in-sample markout yields **−28.7 bp net OOS** (BTC −20 / ETH −36 / SOL −46) — the winner’s curse, quantified.
- **Formal test (walk-forward, 9 rankers × thousands of cells):** best BTC p=0.09, **0 survivors after FDR**.

Every ranking metric we tried failed the same way. We did not rely on one selector:

Selector	Ranked on	OOS verdict
Naive rankings	edge / Sharpe / hit-rate / cum-PnL	inconclusive → blind (MDE 160 bp)
Reliability	t-stat, empirical-Bayes shrunk edge	null (perm p=0.40, 0 FDR)
Robust sweep	9 rankers × 9,668 cells	nothing graduates (best p=0.17)
Behavioral traits	high-vol frac, size CV, add-freq	descriptive only
PnL / entry-timing	realized PnL, timing rank	doesn’t persist copyably
Maker-share strata	taker-purity conditioned	powered ~0 (−0.007)
Per-wallet copy-edge	vs mechanical benchmark	power-gate blind
Day-weighted top-20	monthly rank	basket ~0 net
Nested wallet features	market-state + wallet	incremental ≈ 0

Two earlier selection studies (a 17-metric per-wallet screen and a sealed cross-validated tournament) reached the same conclusion on the same tape — corroborating, not counted here.

But the null is not total. At the population/decile level a weak signal is real and powered (next two sections handle it under strict both-directions rigor — we neither over-null it nor over-carry it).

6. Issues encountered & solutions

A markout study is mostly a fight against measurement artifacts. The load-bearing ones:

A. Stale-candle pricing (prior copy-trade investigation — the cautionary tale that set this study’s pricing). *Provenance: this finding is from the earlier Hyperliquid copy-trade edge investigation (Jun–Jul 2026), not this cohort pipeline; it is why this study prices post-fill on fine candles.* An apparent **+24 to +31 bp** follower edge (6h, 4/4 folds, CI > 0 even at 22 bp cost) was **entirely a pricing artifact**: entry markout was read off the last *closed hourly candle*, 15–30 min before the actual fill, crediting the follower with the pre-fill run-up and the leader’s own impact. The tell was a too-clean lag-decay curve (71 → 64 → 37 → 21). Re-pricing strictly post-fill collapsed the flagship cell **+39.3 → +0.4 bp**; on 15-min candles it went negative. **Fix adopted here**: strictly post-fill, interval-aware next-bar-close pricing on fine candles — the baseline for everything above.

B. Statistical hygiene.

Pitfall	Guard	Outcome
Survivorship	past-only universe, activity gate, MTM open positions	inflated +159 → +40–92 bp/RT removed
Winner’s curse (~10k hypotheses)	held-out means, BH-FDR, config-null	–28.7 bp OOS; 0/75 cells survive
Look-ahead	fully-closed candles, purge+embargo	a backward-peek $z=+3.86$ collapsed
Gross vs net	realistic HL ~5–6 bp round-trip, turnover-weighted	M1 +3.75 → +0.20 bp net
Non-independence	day-block bootstrap + Romano-Wolf	overlapping 8h holds clustered correctly
Over-nulling and over-carrying	point + CI + positive-control MDE + cross-unit sign	both error directions gated equally

C. Bugs we caught in our own pipeline (via adversarial self-audit), each with before → after:

Bug	Effect	Fix
Test-activity in <i>selection</i> ($t_{end} \geq 8$)	inflated M1 basket +6.9 → +3.75; “beats fade” was a leak	train-only freeze, eval-time attrition (99 → 170 wallets)
Test-fitted matched-control bins	biased the trigger test	train-fitted, frozen
James-Stein shrinkage collapse	flagged 100% of wallets “top” in 4/11 months → recurrence $\geq 3\text{mo}$ 686 → 48 , transition 3.2× → 1.09× (n.s.)	raw-rank fallback when $\tau^2=0$
Missing positive-control MDE	an implied null wasn’t earned	added — window is blind below ~11–16 bp
Three conflated “top-decile” sets	ambiguous cohort	one canonical train-only set frozen to disk

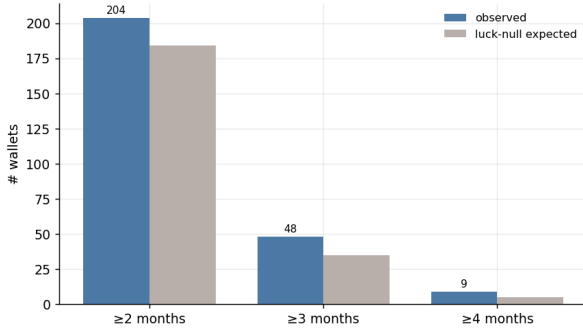
7. The persistence that *is* real → the 121 cohort

Ranking *specific* wallets fails, but **rank persistence at the population level is real and powered** — and this is where over-nulling would have buried a genuine signal:

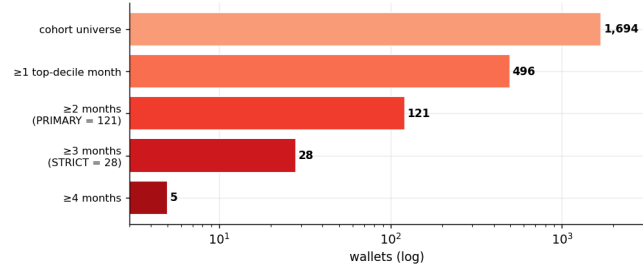
- Adjacent-month **rank-IC 8h = +0.093 [+0.061, +0.126]**, 10/10 folds, **p=0.002** (4h +0.081, 24h +0.085).
- Top-decile **forward markout +5.72 vs field +1.24 = +4.48 bp [+2.17, +7.54]**, 9/10 folds.
- Recurrence beyond luck: wallets top-decile in ≥ 3 months = **48 observed vs 35 by luck (p=0.017)** — the only recurrence threshold that clears.

We freeze the recurring cohort as a single, leak-free, on-disk object: **train-only, day-weighted neut_8h, top-decile in ≥ 2 of 6 training months $\rightarrow$ 121 wallets** (≥ 3 months = 28, strict subset).

Recurrence in the top decile vs luck (only ≥ 3 mo clears: 48 vs 35, $p=0.017$)



Building the 121: train-only recurrence in the neut-8h top decile

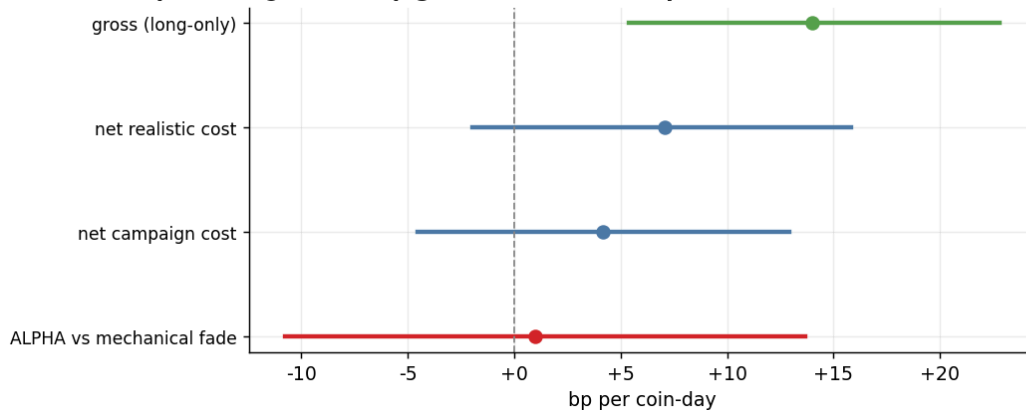


What separates the 121 is **conviction and breadth, not size or activity**: conviction percentile 0.72 vs 0.48 for the field, active in all 4 coins vs 3, near-identical median ticket (\$4.1k vs \$3.9k), and *fewer* entries (268 vs 291).

8. Is the edge skill? The fade decomposition

The 121 cohort's forward markout is real (+4.5 bp) — but is it *wallet skill* or a *market setup* anyone can trade? We benchmark against a costless mechanical fade (`fade_dir = -sign(trailing 8h move)` , priced at the wallets' own entry/exit timestamps so execution cancels):

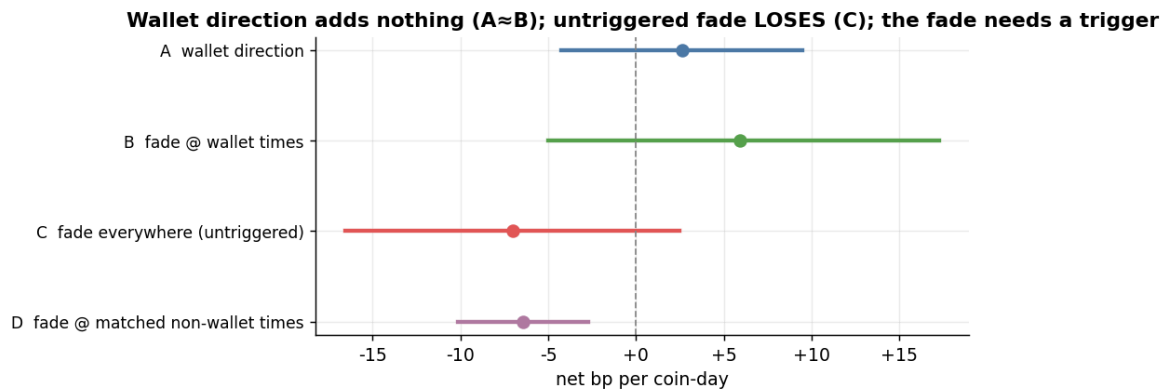
Top-decile @8h: +14bp gross is REAL, but alpha over a costless mechanical fade ≈ 0



Alpha vs fade

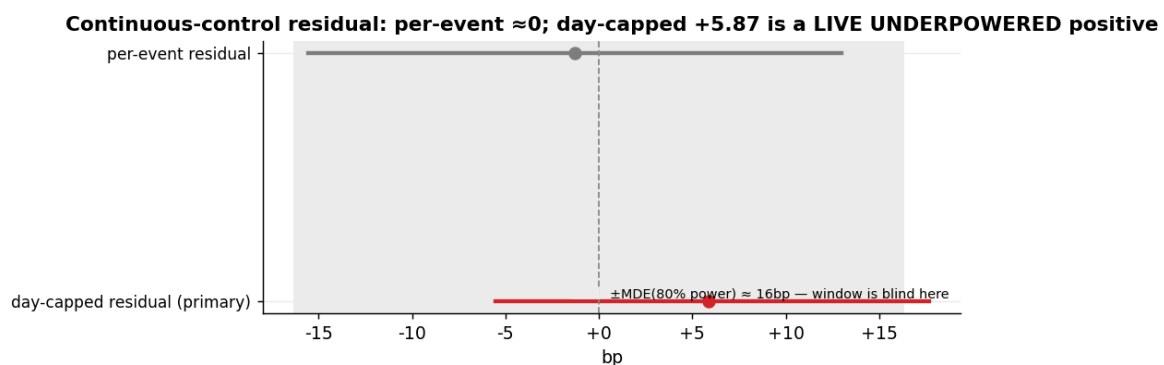
- Top-decile @8h: gross **+13.99 bp** [+5.4, +22.8] is real, net realistic cost **+7.09** — **but alpha over the mechanical fade is +1.00 bp** [-10.75, +13.66], $p=0.44$ (≈ 0). The entire gross edge is the fade.

Decomposing further with a matched-control experiment (A = trade wallet's direction; B = fade at wallet times; C = fade everywhere; D = fade at activity-matched *non-wallet* times):



A/B/C/D

- **Wallet direction adds nothing ($A \approx B$). The untriggered fade loses money net ($C = -7.0$)** — so the fade needs a trigger, but the placebo D fails too, and whether wallet activity is a *uniquely* useful trigger is confounded by wallets trading ~13% larger moves.
- A frozen continuous control (fit $E[\text{fade} \mid \text{trailing move, vol, coin, time}]$ on non-wallet bars, apply to wallet events) resolves the confound: **per-event residual ≈ 0** (the state fully explains the event fade), while the day-capped residual is **+5.87 bp [-5.5, +17.7]** — positive but the window's **MDE is 11–16 bp**, so it is a *live underpowered positive, not a null*.



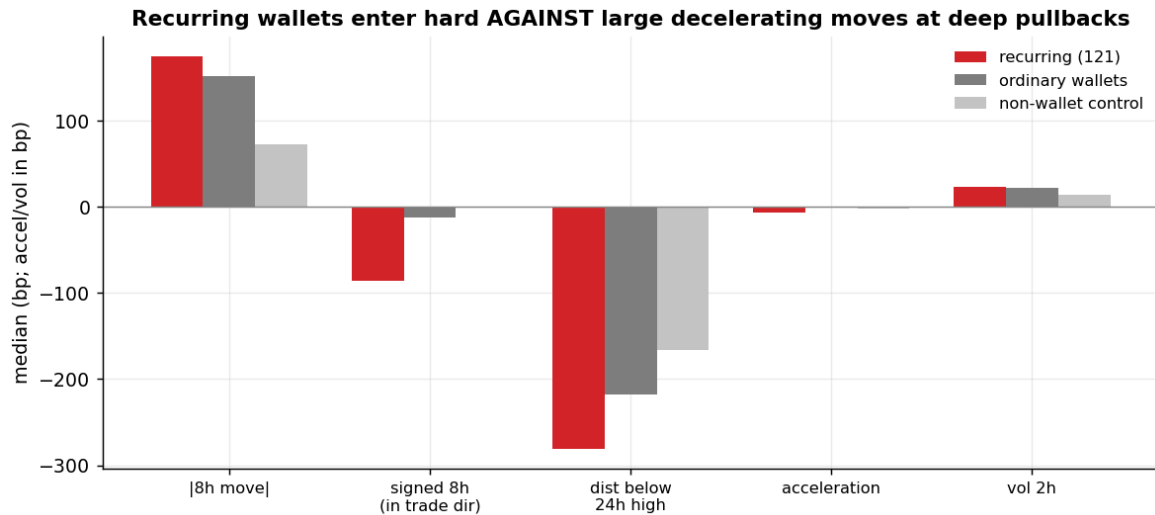
M4 residual + MDE

- Confirming from the other direction: a nested model predicting the 8h return from market state (M_0) vs market state + **every wallet feature** (M_1 : recurring flag, rank, direction, size) gains **-0.0003 cross-fit IC** — wallet identity adds nothing even in-sample.

Verdict (both gates): wallet *ranking* persists (real, powered) but it is *generic short-horizon reversal*; wallet *direction* and *timing* add nothing measurable beyond the observable setup; a small day-capped trigger residual survives as underpowered/unresolved, not as demonstrated skill.

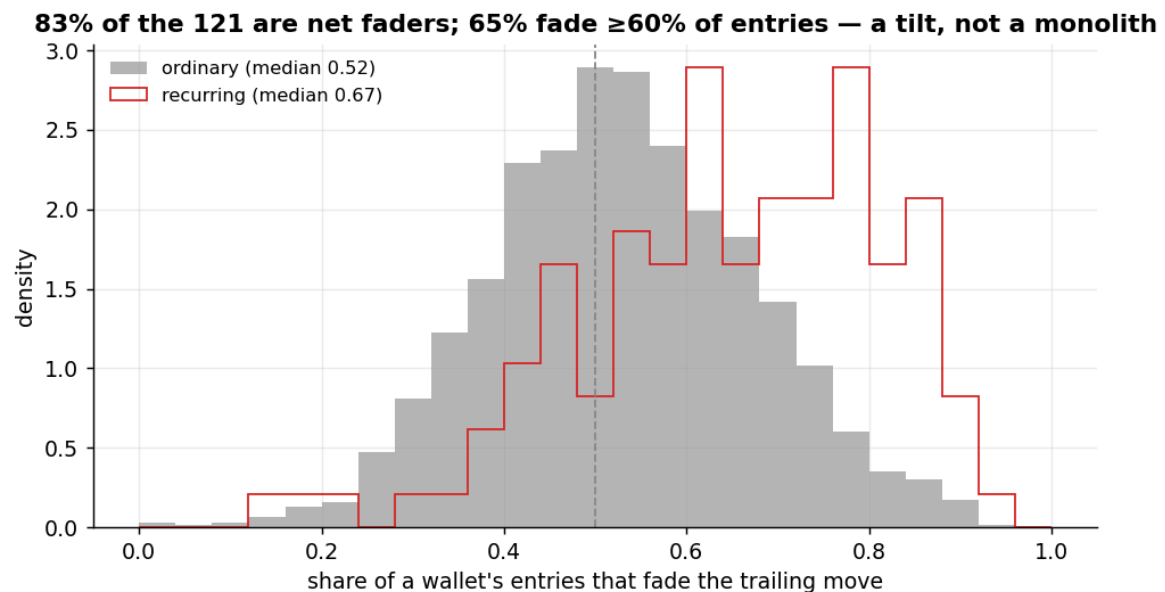
9. Characterizing the 121

The cohort has a clear, single behavioral signature — **aggressive contrarian mean-reversion**:

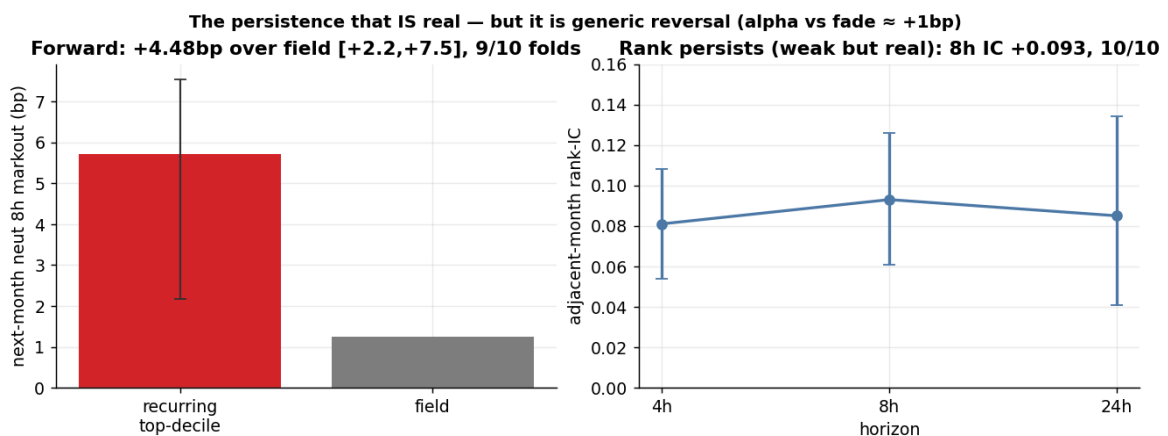


Behavioral decomposition

Recurring wallets enter on larger trailing moves ($|8h|$ 175 vs 151 vs 72 bp for control), **hard against the move** (signed 8h-in-direction -86 vs -12 bp), **deeper below the 24h high** (-281 vs -218 vs -166), as the move **decelerates** (accel -6.2 vs 0), at slightly elevated vol. Per wallet, **83% are net faders, 65% fade $\geq 60\%$ of their entries, 44% $\geq 70\%$** — a strong tilt, $\sim 2\times$ the enrichment of ordinary wallets, but *not a monolith* ($\sim 17\%$ are net momentum/neutral).



Fade fraction



10. What we learned

- **Markout ranking of wallets is not a tradeable selector on this tape.** Nine-plus metrics, all horizons — none persist out-of-sample; selection on in-sample markout costs -28.7 bp. This is a powered, repeatedly-audited negative, not an underpowered shrug.
- **There is a real but weak population-level persistence**, and it is **generic short-horizon reversal**: the recurring top-decile is a set of aggressive contrarian faders whose forward edge is a mechanical fade, fully explained by the size of the preceding move. Wallet identity/direction adds ~ 0 .
- **One question stays open, honestly**: whether wallet *activity* is a uniquely useful *trigger* for that fade — the day-capped residual ($+5.87$ bp) is positive but the historical window is statistically blind below ~ 11 – 16 bp. It is carried as a live underpowered positive, resolvable only with forward data, not mined further here.
- **The methodology is the deliverable as much as the result**: post-fill fine-candle pricing, chronological splits with embargo, day-block inference for overlapping trades, positive-control MDEs before any null, and adversarial self-audit that caught five of our own leaks/bugs.

Reproducibility: cohort in `out/cohort_K_entries.parquet` (frozen recurring set `out/cohort_M_frozen.{txt,parquet}`); figures generated by `src/fig_*.py`; full method + audit trail in `docs/FINDINGS_LEDGER.md`. All OOS numbers are chronological train→test; figures entry-weighted unless labeled.